

EXPERIMENT NO 14

Determination and plotting of Polarization in different Dielectric Medium with their relative permittivity and to find electric susceptibility.

Aim:

To determine and plot polarization in different dielectric medium with their relative permittivity and to find electric susceptibility.

Software required:

- SCILAB version 6.0.1

Program:

```
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
EXP 14.sci x
5 epsilon0 = 8.854e-12;
6
7 //Input values
8 er = input("Enter the relative permittivity (er):-");
9 field = input("Enter the Electric Field (V/m):-");
10
11 // Electric susceptibility
12 chi = er - 1;
13
14 // Polarisation
15 polarization = chi * epsilon0 * field;
16
17 // Display results
18 sprintf("\nElectric Susceptibility (chi) = %.6f\n", chi);
19 sprintf("Polarisation (P) = %.6e C/m^2\n", polarization);
20
21 // Values for plotting
22 x = linspace(0, 10, 50);
23
24 // Susceptibility is constant for a fixed er
25 y = chi * ones(1, length(x));
26
27 // Polarisation as electric field varies
28 E_values = 0.1 * x;
29 s = chi * epsilon0 * E_values;
30
31
32 // Create figure
33 clf();
34
35 // Graph 1: Electric Susceptibility
36 subplot(2, 1, 1);
37 plot(x, y);
38 xlabel("Electric Field (V/m)");
39 ylabel("Electric Susceptibility");
40 title("ELECTRIC SUSCEPTIBILITY");
41
42 // Graph 2: Polarisation
43 subplot(2, 1, 2);
44 plot(E_values, s);
45 xlabel("Electric Field (V/m)");
46 ylabel("Polarisation (C/m^2)");
47 title("POLARIZATION");
48
49
```

Output:

Scilab 2026.1.0 Console

File Edit Control Applications ?

File Browser C:\Users\User\Documents\

Enter the relative permittivity (er): 6×10^{-7}

Enter the Electric Field (V/m): 4×10^{-7}

Electric Susceptibility (χ) = 52.000000
Polarization (P) = $1.519346e-08$ C/m²
exec: Wrong number of output argument(s): 0 expected.

-->

Variable Browser

Name	Value	Type	Visibility	Memory
epsilon0	8.85e-12	Double	local	216 B
epsilon_air	8.85e-12	Double	local	216 B
epsilon_v...	8.85e-12	Double	local	216 B
epsilon_...	7.08e-10	Double	local	216 B
eta_air	377	Double	local	216 B
eta_vacu...	377	Double	local	216 B
eta_water	42.1	Double	local	216 B
mu0	1.26e-06	Double	local	216 B
mu_air	1.26e-06	Double	local	216 B

Command History

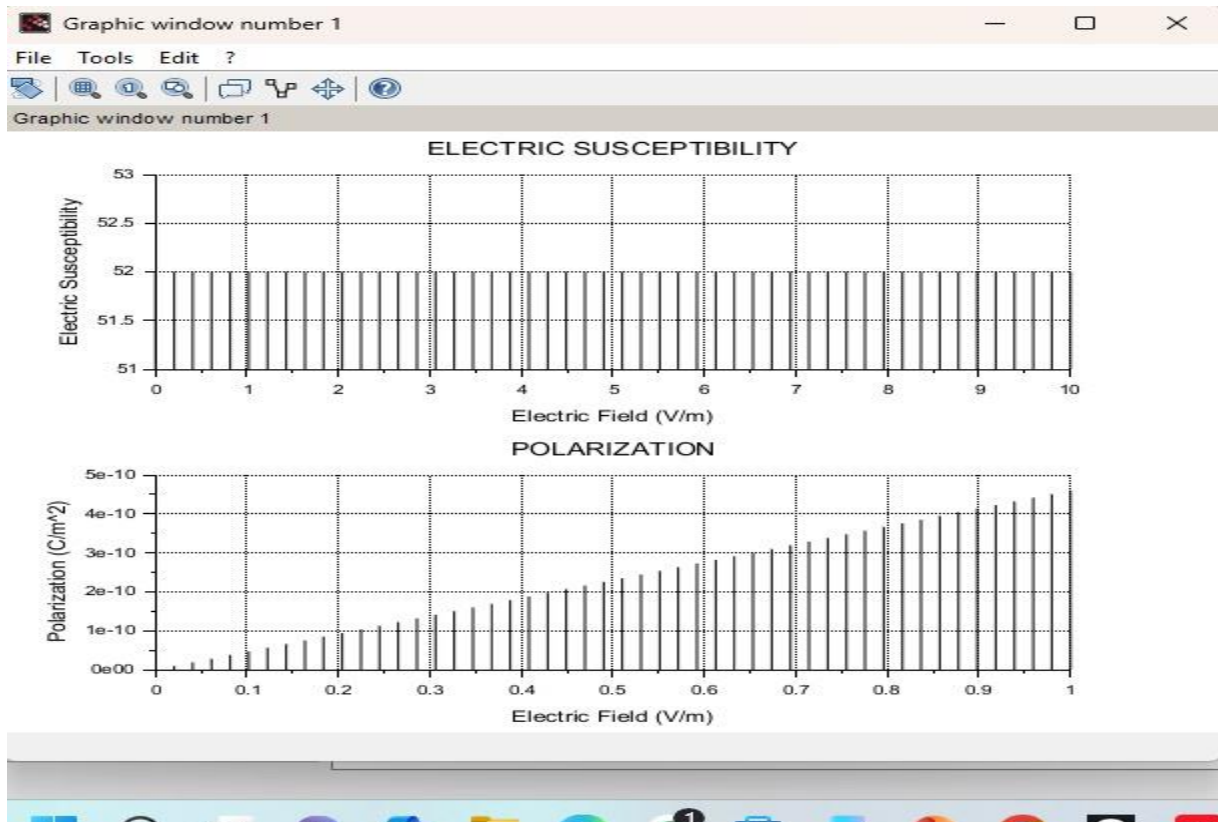
```
// -- 10/08/2026 14:33:17 -- //  
// -- 10/08/2026 14:50:34 -- //  
// -- 24/08/2026 04:27:36 -- //  
// -- 24/08/2026 04:29:27 -- //  
3*10^-7  
6  
-4  
-2  
5  
-3  
-4  
5  
-3  
4
```

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Theoretical calculations:

EXP NO: 14

Determination and plotting of polarization in different Dielectric Medium with their relative permittivity and to find electric susceptibility.

Output:

Enter the ϵ_r value =

$$6 \times 10^{-7}$$

Enter the Electric Field =

$$4 \times 10^{-7}$$

Case 3:

Enter the value of m_r =

$$5 \times 10^{-7}$$

Enter the value of H =

$$5 \text{ cm}$$

Enter the value of length =

$$2 \text{ cm}$$

Enter the value of width =

$$3 \text{ mm}$$

Practical Output:

$$\text{Electric Susceptibility } (\chi_e) = 52.000000$$

$$\text{Polarization } (P) = 1.51934 \times 10^{-2} \text{ C/m}^2$$

Theoretical Calculation:

Relative permittivity, $\epsilon_r = 53$

Electric field, $E = 4 \times 10^7 \text{ V/m}$

Permittivity of free space,

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$$

(1) Electric Susceptibility

$$\chi_e = \epsilon_r - 1$$

$$\chi_e = 52$$

(2) Polarization

$$P = \epsilon_0 \chi_e E$$

$$P = 8.854 \times 10^{-12} \times 52 \times 4 \times 10^7$$

$$P = 1.839 \times 10^{-2} \text{ C/m}^2$$

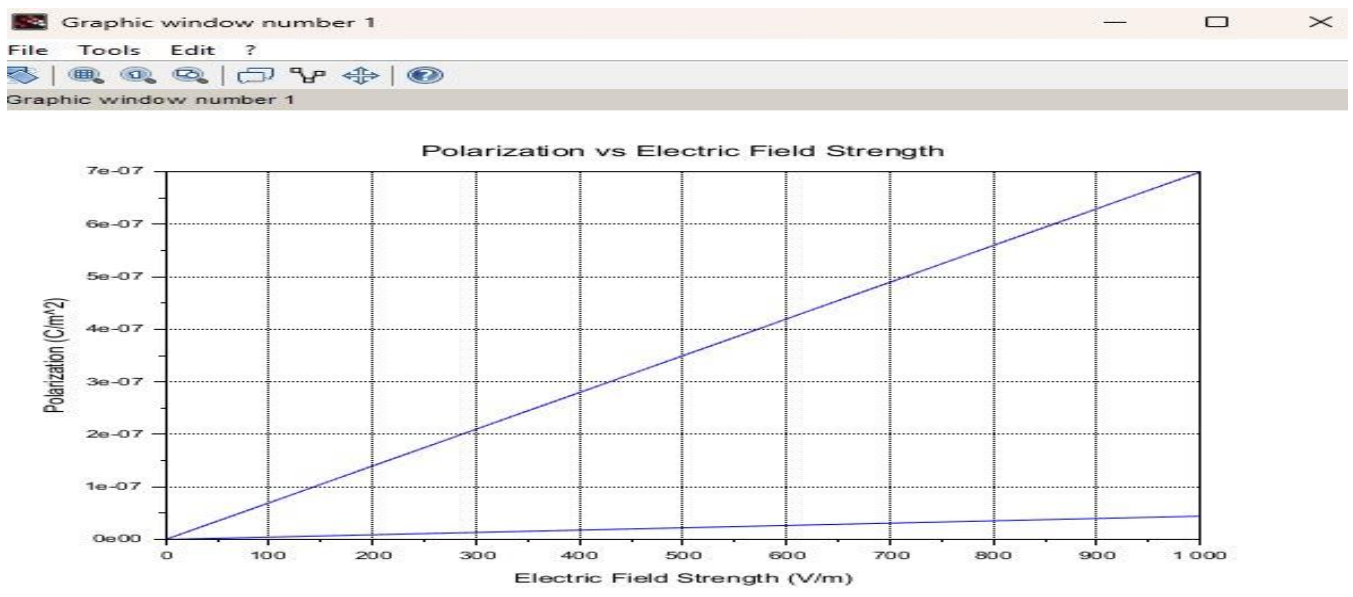
$$P = 1.839632 \times 10^{-2} \text{ C/m}^2$$

Case 1: To determine and plot the polarization and electric susceptibility of electromagnetic waves in three dielectric media: Air, Water, and Glass, using their relative permittivity.

Program:

```
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
EXP 14.sci x
1 clear
2 clear
3
4 // Permittivity of free space
5 epsilon_0 = 8.854e-12;
6
7 // Electric field strength
8 E = linspace(0, 1000, 100);
9
10 // Relative permittivity
11 er_air = 1;
12 er_water = 80;
13 er_glass = 6;
14
15 // Electric susceptibility
16 chi_air = er_air - 1;
17 chi_water = er_water - 1;
18 chi_glass = er_glass - 1;
19
20 // Polarization
21 P_air = epsilon_0 * chi_air * E;
22 P_water = epsilon_0 * chi_water * E;
23 P_glass = epsilon_0 * chi_glass * E;
24
25 // Create graph
26 axis();
27 title();
28
29 // Plot Air
30 plot(E, P_air);
31
32 // Plot Water
33 plot(E, P_water);
34
35 // Plot Glass
36 plot(E, P_glass);
37
38 // Labels
39 xlabel('Electric Field Strength (V/m)');
40 ylabel('Polarization (C/m^2)');
41 title('Polarization vs Electric Field Strength');
42
43 // Grid
44 grid();
45
```

Output:

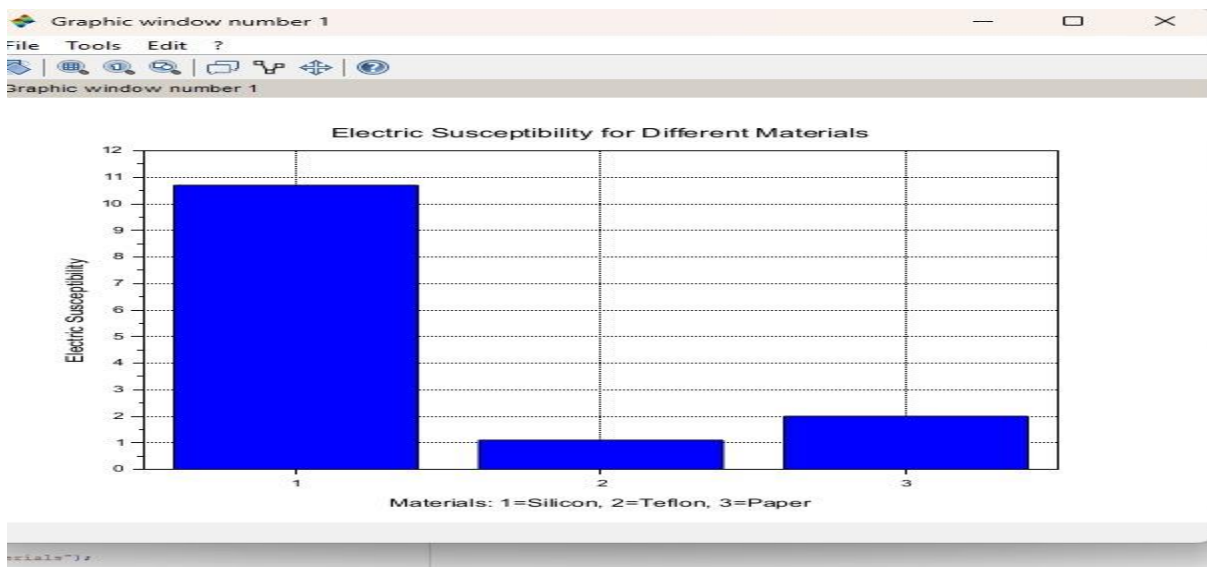


Case 2: To calculate and plot the polarization and electric susceptibility in Silicon, Teflon, and Paper, and understand how their relative permittivity affects the polarization.

Program:

```
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
EXP 14.sci
1 clear;
2 clear;
3
4 // Constant
5 epsilon0 = 8.854e-12;
6
7 // Relative permittivity
8 epsilon_r = [11.7 2.1 3.0];
9
10 // Electric susceptibility
11 chi_e = epsilon_r - 1;
12
13 // Electric field
14 E = 1e6;
15
16 // Polarisation
17 polarization = epsilon0 .* chi_e .* E;
18
19 // Display values
20
21 disp("Electric Susceptibility:");
22 disp(chi_e);
23
24 disp("Polarisation (C/m^2):");
25 disp(polarization);
26
27 // Graph 1:- Electric Susceptibility
28 clf(1);
29
30
31
32 bar(chi_e);
33 xlabel("Materials: 1=Silicon, 2=Teflon, 3=Paper");
34 ylabel("Electric Susceptibility");
35 title("Electric Susceptibility for Different Materials");
36 grid();
37
38 // Graph 2:- Polarisation
39
40
41
42 bar(polarization);
43 xlabel("Materials: 1=Silicon, 2=Teflon, 3=Paper");
44 ylabel("Polarisation (C/m^2)");
45 title("Polarisation for Different Materials");
46 grid();
47
48
```

Output:



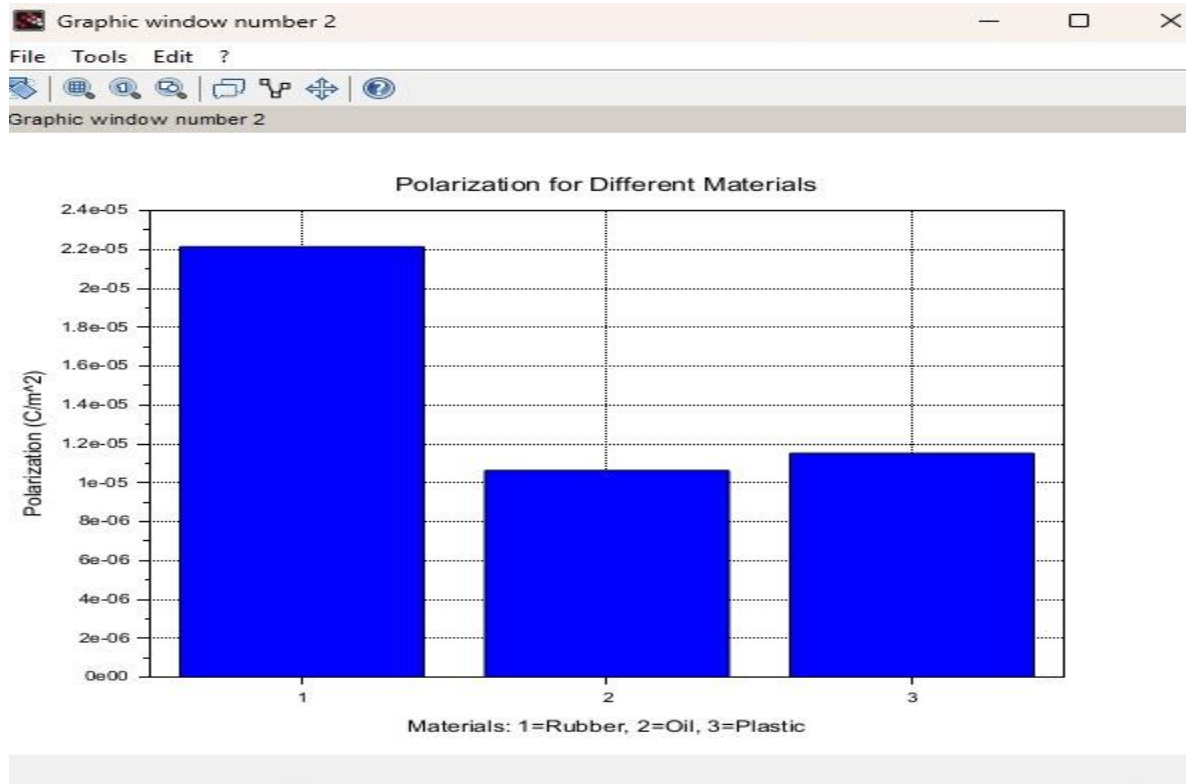
Case 3: To explore how different dielectric materials (Rubber, Oil, and Plastic) affect the polarization and electric susceptibility, and to compare their relative permittivity.

Program:

```
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
EXP 14.sci X

1 clc;
2 clear;
3
4 //Constant
5 epsilon0 = 8.854e-12;
6
7 //Relative permittivity
8 epsilon_r = [2.5 2.2 2.3];
9
10 //Electric susceptibility
11 chi_e = epsilon_r - 1;
12
13 //Applied electric field
14 E = 1e6;
15
16 //Polarisation
17 polarisation = epsilon0 .* chi_e .* E;
18
19
20 //=====
21 //DISPLAY RESULTS
22 //=====
23
24 disp("Electric-Susceptibility:");
25 disp(chi_e);
26
27 disp("Polarisation-(C/m^2):");
28 disp(polarisation);
29
30 //=====
31 //GRAPH-1: ELECTRIC-SUSCEPTIBILITY
32 //=====
33
34 scf(1);
35 clf();
36
37 bar(chi_e);
38
39 xlabel("Materials: -1=Rubber, -2=Oil, -3=Plastic");
40 ylabel("Electric-Susceptibility");
41 title("Electric-Susceptibility-for-Different-Materials");
42
43
44
45
46 //=====
47 //GRAPH-2: POLARISATION
48 //=====
49
50
51 scf(2);
52 clf();
53
54 bar(polarisation);
55
56 xlabel("Materials: -1=Rubber, -2=Oil, -3=Plastic");
57 ylabel("Polarisation-(C/m^2)");
58 title("Polarisation-for-Different-Materials");
59
60 xgrid();
61
```

Output:



Case 1: Examine the effect of length of a ferromagnetic material on the magnetic flux density (B) for the following conditions

- ☐ The material is ferromagnetic with relative permeability (μ_r) of 500.
- ☐ The magnetic field intensity (H) is held constant at 100 A/m.
- ☐ The length of the material varies from 0.1 m to 2 m.

Program:

```
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
EXP 14.sci X
1 clear;
2 clear;
3 clf;
4
5 // Constants
6 mu_r = 800; // Relative permeability
7 H = 100; // Magnetic field intensity (A/m)
8 mu_0 = 4 * pi * 10^-7; // Permeability of free space (H/m)
9
10 // Lengths from 0.1 m to 2 m
11 lengths = 0.1:0.1:2;
12
13 // Initialise array for magnetic flux density
14 B_values = zeros(1, length(lengths));
15
16 // Calculate magnetic flux density
17 for i = 1:length(lengths)
18     B_values(i) = mu_r * mu_0 * H;
19 end
20
21 // Display results
22 disp('Length (m) ----- Magnetic Flux Density (T)');
23
24 for i = 1:length(lengths)
25     fprintf('%1.1f ----- %1.6f\n', lengths(i), B_values(i));
26 end
27
28 // Plot the results
29 plot(lengths, B_values, '-o');
30 xlabel('Length of Ferromagnetic Material (m)');
31 ylabel('Magnetic Flux Density B (T)');
32 title('Effect of Length on Magnetic Flux Density');
33 grid();
34
```

Output:

Scilab 2026.1.0 Console

File Edit Control Applications ?

File Browser

C:\Users\user\Documents\

Documents

My Music

My Pictures

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ex 6.sci

EXP 14.sci~

File/directory filter

Scilab 2026.1.0 Console

"Length (m) Magnetic Flux Density (T)"

0.1	0.062832
0.2	0.062832
0.3	0.062832
0.4	0.062832
0.5	0.062832
0.6	0.062832
0.7	0.062832
0.8	0.062832
0.9	0.062832
1.0	0.062832
1.1	0.062832
1.2	0.062832
1.3	0.062832
1.4	0.062832
1.5	0.062832
1.6	0.062832
1.7	0.062832
1.8	0.062832
1.9	0.062832
2.0	0.062832

-->

Variable Browser

Name	Value	Type	Visibility	Memory
B_values	1x20	Double	local	368 B
H	100	Double	local	216 B
i	20	Double	local	216 B
lengths	1x20	Double	local	368 B
mu_0	1.256-06	Double	local	216 B
mu_r	500	Double	local	216 B

Command History

EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes

File Edit Format Options Window Execute ?

EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes

EXP 14.sci X

```
1 clear;
2 clear;
3 clf;
4
5 // Constants
6 mu_r = 800; // Relative permeability
7 H = 100; // Magnetic field intensity (A/m)
8 mu_0 = 4 * pi * 10^-7; // Permeability of free space (H/m)
9
10 // Lengths from 0.1 m to 2 m
11 lengths = 0.1:0.1:2;
12
13 // Initialise array for magnetic flux density
14 B_values = zeros(1, length(lengths));
15
16 // Calculate magnetic flux density
17 for i = 1:length(lengths)
18     B_values(i) = mu_r * mu_0 * H;
19 end
20
21 // Display results
22 disp('Length (m) ----- Magnetic Flux Density (T)');
23
24 for i = 1:length(lengths)
25     fprintf('%1.1f ----- %1.6f\n', lengths(i), B_values(i));
26 end
27
28 // Plot the results
29 plot(lengths, B_values, '-o');
30 xlabel('Length of Ferromagnetic Material (m)');
31 ylabel('Magnetic Flux Density B (T)');
32 title('Effect of Length on Magnetic Flux Density');
33 grid();
34
```

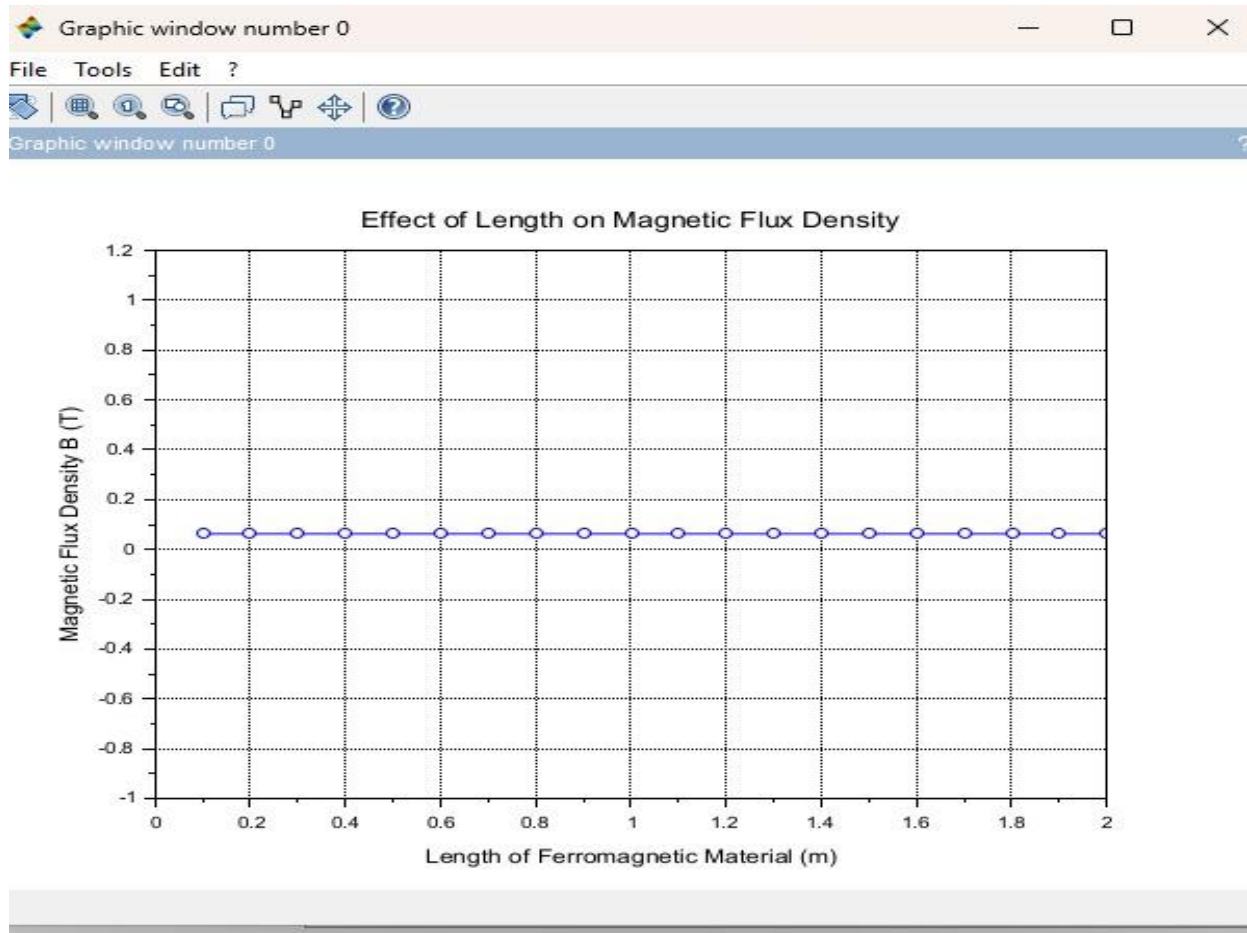
Snipping Tool

Screenshot copied to clipboard

Automatically saved to screenshots folder.

Markup and share

11:29 AM 8/30/2026



Case 2: To investigate how the magnetic flux density (B) in a ferromagnetic material changes with different widths at a constant length and magnetic field intensity.

Conditions:

- Material: Soft iron (same as in Case Study 1)
- Length (L): Constant at 10 cm
- Magnetic Field Intensity (H): Constant at 1000 A/m
- Width (W): Varies from 2 cm to 10 cm (i.e., 2 cm, 4 cm, 6 cm, 8 cm, 10 cm)

Program:

```
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
EXP 14.sci x
1 clc;
2 clear;
3 clf;
4
5 //Constants
6 L = 0.1; //Length in meters (10 cm)
7 H = 1000; //Magnetic field intensity (A/m)
8 mu_0 = 4*pi*10^(-7); //Permeability of free space (H/m)
9
10 //Widths in meters (2 cm to 10 cm)
11 widths = [0.02, 0.04, 0.06, 0.08, 0.10];
12 num_widths = length(widths);
13
14 //Relative permeability of soft iron
15 mu_r = 1000;
16
17 //Initialise array
18 B_values = zeros(1, num_widths);
19
20 //Calculate magnetic flux density
21 for i = 1:num_widths
22     B_values(i) = mu_0 * mu_r * H;
23 end
24
25 //Display results
26 disp("Width (m) ---- Flux Density (T)");
27
28 for i = 1:num_widths
29     fprintf("%e.2f-----%e\n", widths(i), B_values(i));
30 end
31
32 //Plot the results
33 plot(widths, B_values, '-o');
34
35 xlabel("Width of Ferromagnetic Material (m)");
36 ylabel("Magnetic Flux Density B (T)");
37 title("Effect of Width on Magnetic Flux Density");
38
39 xgrid();
40
```

Output:

Scilab 2026.1.0 Console

File Edit Control Applications ?

File Browser

C:\Users\user\Documents\

Documents

My Music

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ex 6.sd

EXP 14.sd

File/directory filter

Width (m) Flux Density (T)

0.02	1.256637
0.04	1.256637
0.06	1.256637
0.08	1.256637
0.10	1.256637

--> |

Variable Browser

Name	Value	Type	Visibility	Memory
B_values	1x5	Double	local	248 B
H	1e+03	Double	local	216 B
L	0.1	Double	local	216 B
mu_0	1.25e-06	Double	local	216 B
mu_r	1e+03	Double	local	216 B
num_widths	5	Double	local	216 B
widths	1x5	Double	local	248 B

Command History

EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes

File Edit Format Options Window Execute ?

EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes

EXP 14.sci x

```
1 clc;
2 clear;
3 clf;
4
5 //Constants
6 L = 0.1; //Length in meters (10 cm)
7 H = 1000; //Magnetic field intensity (A/m)
8 mu_0 = 4*pi*10^(-7); //Permeability of free space (H/m)
9
```

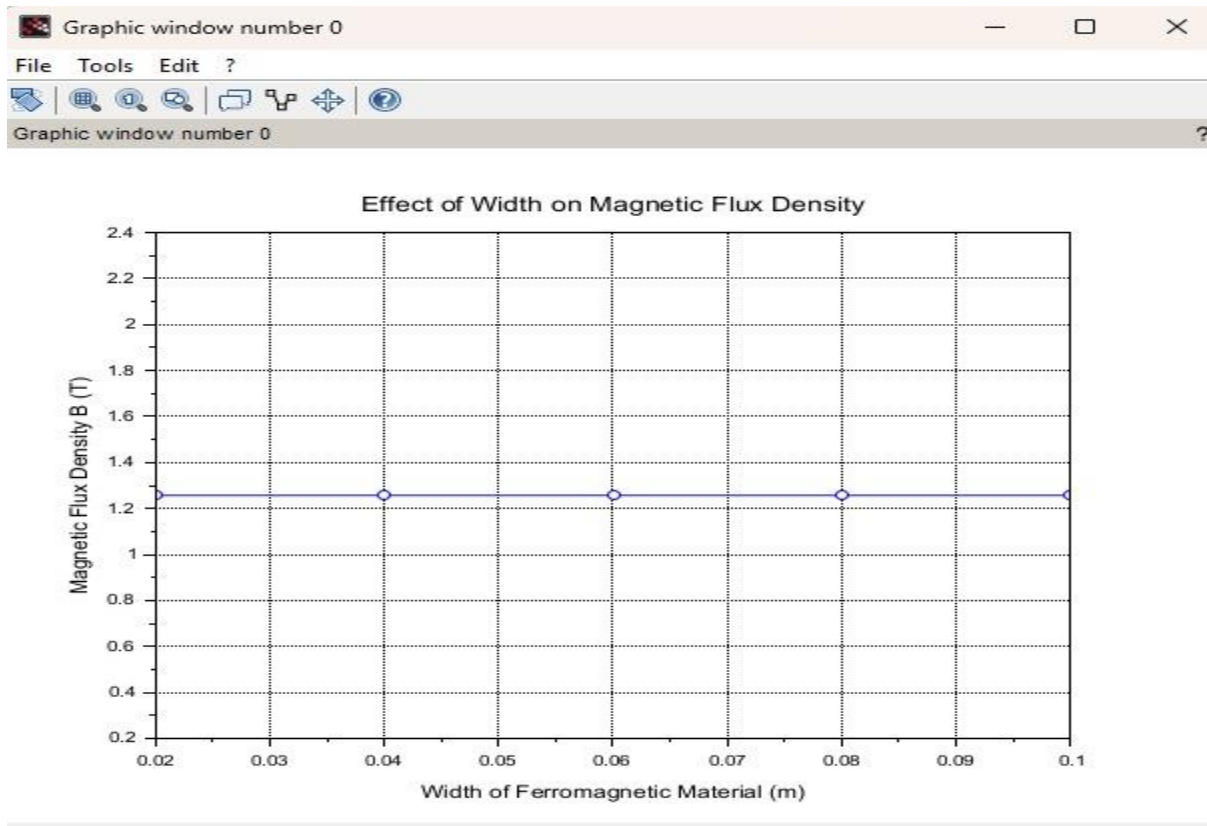
Snipping Tool

Screenshot copied to clipboard

Automatically saved to screenshots folder.

Markup and share

11:31 AM 8/30/2026



Case 3: To examine how the magnetic flux density (B) in a ferromagnetic material changes with different values of magnetic field intensity at constant length and width.

Conditions:

- Material: Soft iron (same as in the previous studies)
- Length (L): Constant at 10 cm
- Width (W): Constant at 5 cm
- Magnetic Field Intensity (H): Varies from 200 A/m to 2000 A/m (i.e., 200 A/m, 500 A/m, 1000 A/m, 1500 A/m, 2000 A/m)

Program:

```
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes
EXP 14.sci X
1 clc;
2 clear;
3 clf();
4
5 // Define constants
6 L = 0.1; // Length in meters (10 cm)
7 W = 0.05; // Width in meters (5 cm)
8
9 H_values = [200, 500, 1000, 1500, 2000]; // H in A/m
10
11 mu_0 = 4 * pi * 10^(-7); // Permeability of free space (H/m)
12 mu_r = 1000; // Relative permeability of soft iron
13
14 // Initialise array for magnetic flux density
15 B_values = zeros(1, length(H_values));
16
17 // Calculate B for each H value
18 for i = 1:length(H_values)
19     H = H_values(i);
20     mu = mu_0 * mu_r; // Absolute permeability
21     B_values(i) = mu * H; // B = mu * H
22 end
23
24 // Display results
25 disp("Magnetic Field Intensity (A/m) --- Magnetic Flux Density (T)");
26
27 for i = 1:length(H_values)
28     fprintf("%10.0f-----%10.6f\n", H_values(i), B_values(i));
29 end
30
31 // Plot the results
32 plot(H_values, B_values, "-o");
33
34 xlabel("Magnetic Field Intensity H (A/m)");
35 ylabel("Magnetic Flux Density B (T)");
36 title("Magnetic Flux Density vs Magnetic Field Intensity for Soft Iron");
37
38 grid();
39
```

Output:

SciLab 2026.1.0 Console

File Edit Control Applications ?

File Browser: C:\Users\user\Documents\EXP 14.sci

Console Output:

```
"Magnetic Field Intensity (A/m)   Magnetic Flux Density (T)"
200                               0.251327
500                               0.628319
1000                              1.256637
1500                              1.884956
2000                              2.513274
```

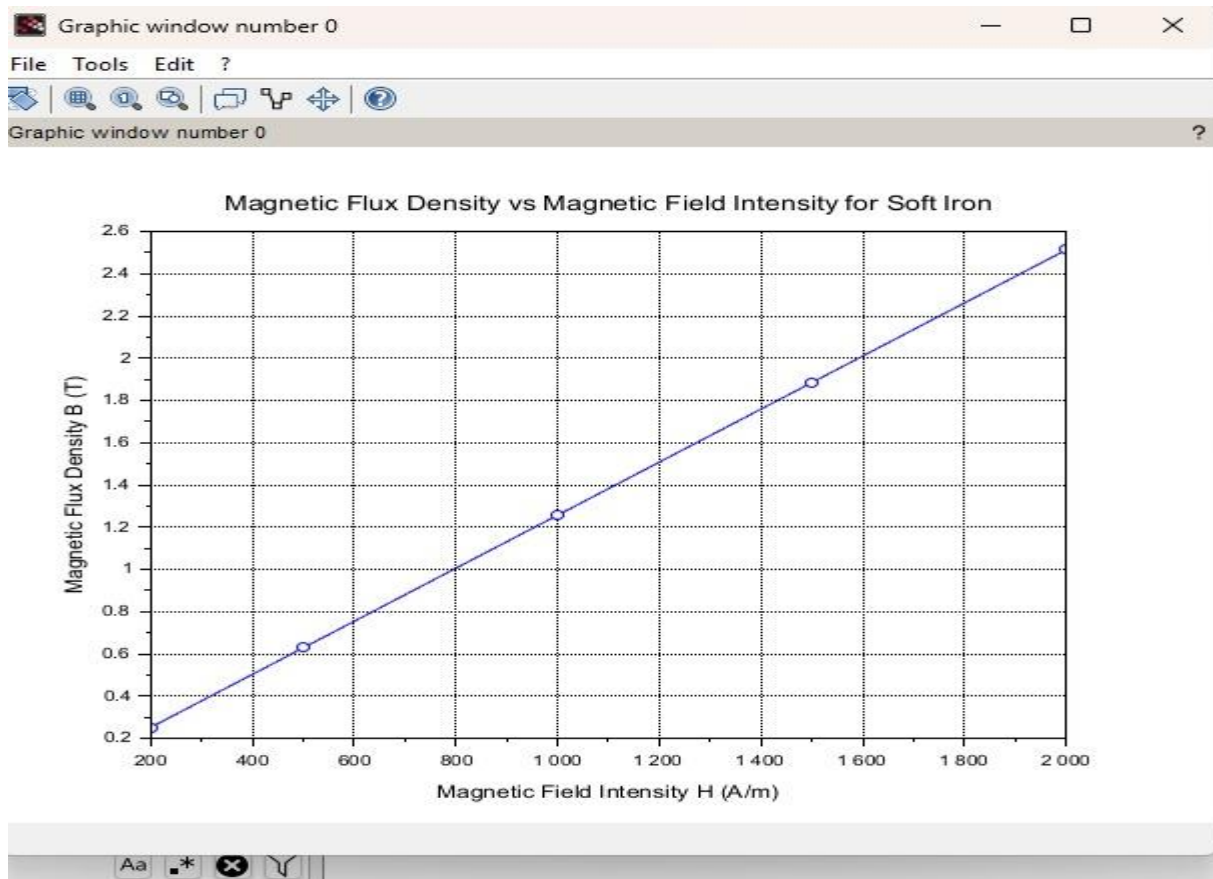
Variable Browser:

Name	Value	Type	Visibility	Memory
B_values	1x5	Double	local	248 B
H	2e+03	Double	local	216 B
H_values	1x5	Double	local	248 B
L	0.1	Double	local	216 B
W	0.05	Double	local	216 B
i	5	Double	local	216 B
mu	0.001256	Double	local	216 B
mu_0	1.256e-06	Double	local	216 B
mu_r	1e+03	Double	local	216 B

Command History:

```
4
2
5
3
4
5
```

Snipping Tool: Screenshot copied to clipboard. Automatically saved to screenshots folder.



Result : Thus the program was verified for program for Magnetic flux density in ferromagnetic materials using SCILAB was successfully.